

CS5340 - Uncertainty Modelling in AI

Lecture 7: Mixture Models and the EM Algorithm

Scribed by: Claude (Anthropic) | Curated By: Assoc. Prof. Lee Gim Hee
Semester 1, AY2026/27 — National University of Singapore

Disclaimer: This scribe was generated by **Claude (Anthropic)** and curated by **Assoc. Prof. Lee Gim Hee**. It is a **supplementary revision aid** that summarizes the lecture slides, which remain the **primary learning material** prepared by the lecturer. Despite curation, the scribe may still contain errors, omissions, or simplifications. Please use it only alongside the official slides, textbook references, and your own lecture notes.

1 Lecture Overview

- **Motivation:** in Lecture 6 every random variable was observed (**complete data**), and the DGM log-likelihood decomposed into closed-form local problems. This lecture asks what happens when some variables are **never observed** – the **latent variables** – and we must learn the parameters from **incomplete data**.
- **Running example:** clustering. Given points $\mathbf{x}_1, \dots, \mathbf{x}_N$ we must find both the **parameters of the clusters** and **which cluster each point belongs to** – a chicken-and-egg problem.
- **Roadmap:**
 1. **K-means:** a non-probabilistic approach that alternates hard assignments and centre updates.
 2. **Gaussian mixture models (GMMs):** a probabilistic model with a 1-of- K latent variable $\mathbf{z}$, the responsibilities $\gamma(z_k)$, and why maximum likelihood is **no longer closed-form**.
 3. **EM for GMMs:** the iterative E-step / M-step scheme suggested by the ML equations.
 4. **The general EM algorithm:** the expected complete-data log-likelihood $Q(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$, singularities and MAP.
 5. **Theory:** the decomposition $\ln p(\mathbf{X} | \boldsymbol{\theta}) = \mathcal{L}(q, \boldsymbol{\theta}) + \text{KL}(q||p)$ and why EM never decreases the likelihood.

Learning Outcomes

By the end of this lecture, you should be able to:

1. Use the non-probabilistic **k-means algorithm** to solve the clustering problem.
2. Describe the **Gaussian-mixture model**.
3. Apply the **Expectation-Maximization algorithm** for estimation of both the unknown parameters and latent variables.

2 Motivation: Clusters and Data Association

Suppose we are given many images of handwritten digits (MNIST), and we extract a 2D representation $\mathbf{x}_n \in \mathbb{R}^2$ of each image. Plotting $\mathbf{x}_1, \dots, \mathbf{x}_N$ reveals **clusters**, and it turns out that points of the **same digit tend to fall in the same cluster**. This suggests that **clusters can be used to represent complex distributions**.

Key Difficulty: The Chicken-and-Egg Problem

1. How do we find the **unknown parameters** of the clusters?
2. Which cluster does each data point belong to, i.e. **data association**?

If we knew the assignments, estimating each cluster's parameters would be easy (Lecture 6). If we knew the cluster parameters, assigning each point would be easy. We know **neither**, which is what makes the problem hard – and what motivates **alternating** between the two.

3 Non-Probabilistic Approach: K-Means**Definition : The Clustering Problem**

Given: (1) a data set $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ of observations; (2) the number of clusters K .

Find: (1) the K **cluster centres** $\{\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K\}$, i.e. the unknown parameters (assuming each cluster is a circle); (2) the **assignment** of each point $\mathbf{x}_n$ to a cluster centre k , i.e. the data association.

Definition : 1-of- K Coding and the Distortion Measure

For each data point $\mathbf{x}_n$ we introduce a set of **binary indicator variables**

$$r_{nk} \in \{0, 1\} \quad \forall k = 1, \dots, K, \quad \text{s.t.} \quad \underbrace{\sum_k r_{nk}}_{\text{1-of-}K \text{ constraint}} = 1,$$

describing which of the K clusters $\mathbf{x}_n$ is assigned to; the constraint ensures each point is assigned to **only one** cluster. The **goal** of k-means is to find $\{r_{nk}\}$ and $\{\boldsymbol{\mu}_k\}$ that minimize the **distortion measure**

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2, \quad \text{s.t.} \quad \sum_k r_{nk} = 1,$$

i.e. the **sum-of-squares** of the distances of each point to its assigned centre.

Key Rule: The Two Steps of K-Means

Assignment step (minimize J w.r.t. $\{r_{nk}\}$, $\{\boldsymbol{\mu}_k\}$ fixed). Each $\mathbf{r}_n$ can be optimized **separately** since they are independent:

$$\arg \min_{\mathbf{r}_n} \sum_k r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2 \quad \text{s.t.} \quad \sum_k r_{nk} = 1 \quad \Rightarrow \quad r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2 \\ 0 & \text{otherwise.} \end{cases}$$

By inspection, the minimum assigns $\mathbf{x}_n$ to the **current closest cluster centre**.

Update step (minimize J w.r.t. $\{\boldsymbol{\mu}_k\}$, $\{r_{nk}\}$ fixed). Each $\boldsymbol{\mu}_k$ can be optimized **separately**. Differentiating $L = \sum_n r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$ w.r.t. $\boldsymbol{\mu}_k$ and equating to zero,

$$2 \sum_{n=1}^N r_{nk} (\mathbf{x}_n - \boldsymbol{\mu}_k) = 0 \quad \Rightarrow \quad \boldsymbol{\mu}_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}}$$

the **mean of all points assigned to cluster k** – hence “k-means”!

Algorithm: K-Means (a.k.a. Lloyd's Algorithm)

- 1: **Initialization:** randomly choose initial values for $\{\boldsymbol{\mu}_k\}$
- 2: **repeat**
- 3: **Assignment step:** $r_{nk} \leftarrow 1$ if $k = \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2$, else 0 $\triangleright$ minimize J w.r.t. $\{r_{nk}\}$
- 4: **Update step:** $\boldsymbol{\mu}_k \leftarrow \sum_n r_{nk} \mathbf{x}_n / \sum_n r_{nk}$ $\triangleright$ minimize J w.r.t. $\{\boldsymbol{\mu}_k\}$
- 5: **until** converged

Each step can only **decrease** (or keep) J , and there are finitely many assignments, so the algorithm converges – though possibly to a **local** minimum that depends on the initialization. In the slides' example ($K = 2$, Old Faithful data, from Bishop) J drops from about 1000 to under 100 and the algorithm has converged after the third update step.

Worked Example: Image Segmentation

Treat each pixel's colour as a point in RGB space and run k-means with K clusters; then replace every pixel by the colour of its cluster centre $\boldsymbol{\mu}_k$. With $K = 2$ or 3 the image becomes a crude poster of a few flat colours; with $K = 10$ it is already close to the original. This is also a form of **lossy compression**: each pixel only needs to store the index k of its centre. (Image source: C. Bishop, "Pattern Recognition and Machine Learning".)

4 Probabilistic Approach: Gaussian Mixture Models

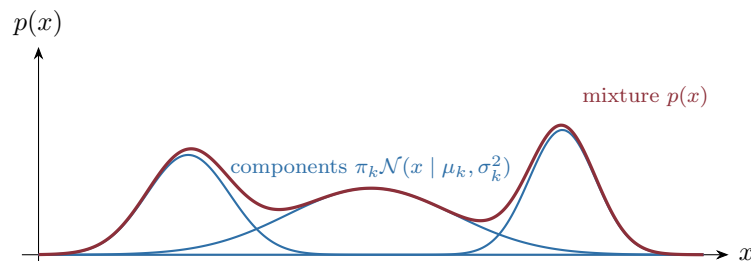
Can we model the clustering problem with a probabilistic approach? The “Old Faithful” data set (272 eruption measurements of the Old Faithful geyser, Yellowstone) forms **two dominant clumps**: a single Gaussian cannot capture this structure, but a **linear superposition of two Gaussians** characterizes it well.

Definition : Gaussian Mixture Model (GMM)

Mixture distributions are probabilistic models formed by taking linear combinations of more basic distributions. With Gaussians, this is a **Gaussian mixture model**, a superposition of K Gaussian densities:

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad 0 \leq \pi_k \leq 1, \quad \sum_{k=1}^K \pi_k = 1.$$

- Each $\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is a **component** of the mixture, with its own mean $\boldsymbol{\mu}_k$ and covariance $\boldsymbol{\Sigma}_k$.
- The π_k are the **mixing coefficients**.
- A linear combination of a sufficient number of Gaussians can **approximate almost any continuous density** with arbitrary accuracy.



A 1D GMM (red) formed by 3 Gaussians (blue). (Schematic redrawn by the scribes after Bishop.)

4.1 The Latent-Variable View

Definition : The 1-of- K Latent Variable $\mathbf{z}$

Introduce a K -dimensional **binary random variable** $\mathbf{z}$ with a **1-of- K representation**: $z_k = 1 \Rightarrow z_{j \neq k} = 0$ indicates the assignment of $\mathbf{x}$ to the k th Gaussian. Its values satisfy $z_k \in \{0, 1\}$ and $\sum_k z_k = 1$, so there are K possible states. Then:

- **Marginal of $\mathbf{z}$** – a categorical distribution in terms of the mixing coefficients:

$$p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k} = \text{Cat}_{\mathbf{z}}[\boldsymbol{\pi}], \quad \boldsymbol{\pi} = [\pi_1, \dots, \pi_K], \quad 0 \leq \pi_k \leq 1, \quad \sum_k \pi_k = 1.$$

- **Conditional of $\mathbf{x}$ given $\mathbf{z}$** – a Gaussian:

$$p(\mathbf{x} \mid z_k = 1) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad \Longleftrightarrow \quad p(\mathbf{x} \mid \mathbf{z}) = \prod_{k=1}^K \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k}.$$

- **Joint** – given by the two-node DGM $\mathbf{z} \rightarrow \mathbf{x}$: $p(\mathbf{x}, \mathbf{z}) = p(\mathbf{z}) p(\mathbf{x} \mid \mathbf{z})$.

Proof: Marginalizing $\mathbf{z}$ Recovers the GMM

Summing the joint over all K states of the latent variable,

$$\begin{aligned} p(\mathbf{x}) &= \sum_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x} \mid \mathbf{z}) = \sum_{\mathbf{z}} \prod_{k=1}^K \pi_k^{z_k} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k} \\ &= \underbrace{\left(\prod_k \pi_k^{z_k} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k} \right)_{z_1=1}}_{\pi_1 \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)} + \dots + \underbrace{\left(\prod_k \pi_k^{z_k} \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k} \right)_{z_K=1}}_{\pi_K \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_K)} \\ &= \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \end{aligned}$$

because in the state $z_j = 1$ every factor with $k \neq j$ is raised to the power 0. ■

So a GMM is exactly the marginal of a DGM with a hidden discrete “which component” variable. Having an explicit $\mathbf{z}$ is what lets us reason about data association probabilistically.

Definition : Responsibility $\gamma(z_k)$

The posterior probability of component k given $\mathbf{x}$, found using **Bayes’ theorem**:

$$\gamma(z_k) \equiv p(z_k = 1 \mid \mathbf{x}) = \frac{p(z_k = 1) p(\mathbf{x} \mid z_k = 1)}{\sum_{j=1}^K p(z_j = 1) p(\mathbf{x} \mid z_j = 1)} = \frac{\pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

π_k is the **prior** probability of $z_k = 1$, and $\gamma(z_k)$ the corresponding **posterior** once $\mathbf{x}$ is observed. $\gamma(z_k)$ is the **responsibility** that component k takes for “explaining” the observation $\mathbf{x}$ – a **soft** version of k-means’ hard r_{nk} .

5 Maximum Log-Likelihood for GMMs

Definition : The Incomplete-Data Log-Likelihood

For N i.i.d. observations $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, $\mathbf{x}_n \in \mathbb{R}^D$, each with its own latent $\mathbf{z}_n$:

$$\begin{aligned} \ln p(\mathbf{x}_1, \dots, \mathbf{x}_N | \boldsymbol{\theta}) &= \sum_{n=1}^N \ln \sum_{\mathbf{z}_n} p(\mathbf{x}_n, \mathbf{z}_n | \boldsymbol{\theta}) = \sum_{n=1}^N \ln \sum_{\mathbf{z}_n} p(\mathbf{z}_n | \boldsymbol{\pi}) p(\mathbf{x}_n | \mathbf{z}_n, \boldsymbol{\mu}, \boldsymbol{\Sigma}) \\ &= \sum_{n=1}^N \ln \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \boldsymbol{\theta} = \{\pi_1 \dots \pi_K, \boldsymbol{\mu}_1 \dots \boldsymbol{\mu}_K, \boldsymbol{\Sigma}_1 \dots \boldsymbol{\Sigma}_K\}. \end{aligned}$$

This is **incomplete data** because $\mathbf{Z} = \{\mathbf{z}_1, \dots, \mathbf{z}_N\} \in \mathbb{R}^{N \times K}$ is a latent variable.

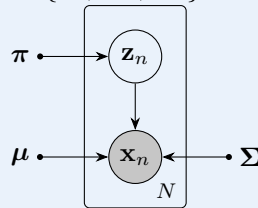


Plate notation: $\mathbf{x}_n$ is observed (shaded), $\mathbf{z}_n$ is hidden. (After Bishop.)

Key Difficulty: A Sum Inside the Logarithm

$$\arg \max_{\boldsymbol{\theta}} \ln p(\mathbf{x}_1, \dots, \mathbf{x}_N | \boldsymbol{\theta}) = \arg \max_{\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}} \sum_{n=1}^N \ln \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

The **summation inside the logarithm** prevents the log from acting directly on the Gaussian. Compare Lecture 6, where $\ln \mathcal{N}$ turned into a quadratic: here we will **no longer obtain a closed-form solution** by setting derivatives to zero.

Worked Example: Stationary Conditions for $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$ and π_k

Means. Setting the derivative of $\ln p(\mathbf{x}_1, \dots, \mathbf{x}_N | \boldsymbol{\theta})$ w.r.t. $\boldsymbol{\mu}_k$ to zero,

$$0 = \sum_{n=1}^N \underbrace{\frac{\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}}_{\gamma(z_{nk})} \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_n - \boldsymbol{\mu}_k).$$

Multiplying by $\boldsymbol{\Sigma}_k$ (assumed non-singular) and rearranging,

$$\boxed{\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n}, \quad N_k = \sum_{n=1}^N \gamma(z_{nk}).$$

(The slide carries a leading minus sign, as in Bishop eq. 9.16; since the right-hand side equals zero, the sign does not affect the result.)

Covariances. Setting the derivative w.r.t. Σ_k to zero,

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^T$$

Mixing coefficients. Maximize w.r.t. π_k subject to $\sum_k \pi_k = 1$ via the auxiliary function with a **Lagrange multiplier** λ :

$$\ln p(\mathbf{X} \mid \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right) \implies 0 = \sum_{n=1}^N \underbrace{\frac{\mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}}_{= N_k / \pi_k} + \lambda.$$

Multiplying both sides by π_k and summing over k using $\sum_k \pi_k = 1$:

$$\sum_{n=1}^N \underbrace{\frac{\sum_k \pi_k \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}}_{=1} = -\lambda \underbrace{\sum_k \pi_k}_{=1} \implies \lambda = -N, \quad \text{so} \quad \boxed{\pi_k = \frac{N_k}{N}}$$

the **average responsibility** which the k th component takes for explaining the data points.

Key Rule: Not Closed-Form, but an Iterative Scheme

All three results look like the complete-data estimates of Lecture 6, with the hard counts replaced by **soft counts** $\gamma(z_{nk})$: N_k is the **effective number of points** assigned to component k , $\boldsymbol{\mu}_k$ is a responsibility-weighted mean, and $\boldsymbol{\Sigma}_k$ a responsibility-weighted covariance.

However, they **do not constitute a closed-form solution**, because the responsibilities $\gamma(z_{nk})$ are themselves functions of $\pi_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$. They do suggest a **simple iterative scheme**: fix the parameters and compute γ ; fix γ and recompute the parameters; repeat. That is the EM algorithm.

6 EM for Gaussian Mixtures

Algorithm: EM for Gaussian Mixtures

Goal: maximize the likelihood w.r.t. $\theta = \{\pi_k, \mu_k, \Sigma_k\}$.

1: **Initialize** μ_k, Σ_k, π_k , and evaluate the initial log-likelihood

2: **repeat**

3: **E-step:** evaluate the responsibilities with the current parameters $\triangleright \gamma(\mathbf{Z})$ is an $N \times K$ table

$$\gamma(z_{nk}) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \mu_j, \Sigma_j)}$$

4: **M-step:** re-estimate the parameters with the current responsibilities, $N_k = \sum_{n=1}^N \gamma(z_{nk})$:

$$\begin{aligned} \mu_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n, & \Sigma_k^{\text{new}} &= \frac{1}{N_k} \sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k^{\text{new}})(\mathbf{x}_n - \mu_k^{\text{new}})^T, \\ \pi_k^{\text{new}} &= \frac{N_k}{N} \end{aligned}$$

5: **Evaluate** the log-likelihood $\ln p(\mathbf{X} | \mu, \Sigma, \pi) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$

6: **until** the parameters or the log-likelihood have converged

Note that Σ_k^{new} uses the **new** mean μ_k^{new} computed in the same M-step.

Worked Example: Illustration on Old Faithful (Bishop)

Initialization: random μ_k , identity Σ_k and $\pi_k = 0.5$ (two circles). **First E-step:** each point is coloured by its posterior – points between the clusters are purple, i.e. **shared** between the two components. **M-steps** at $L = 1, 2, 5, 20$ iterations: the two Gaussians stretch, rotate and move until, at $L = 20$, each tightly covers one clump and almost every point is fully red or fully blue.

Worked Example: One EM Iteration by Hand (Scribe's Example)

Not from the slides; added to make the updates concrete. Take 1D data $x = \{0, 1, 3, 4\}$, $K = 2$, and initialize $\mu_1 = 1, \mu_2 = 3, \sigma_1^2 = \sigma_2^2 = 1, \pi_1 = \pi_2 = 0.5$. The initial log-likelihood is -7.158 .

E-step. With equal π_k and σ_k^2 , the mixing weights and Gaussian normalizers cancel, leaving

$$\gamma(z_{n1}) = \frac{e^{-(x_n-1)^2/2}}{e^{-(x_n-1)^2/2} + e^{-(x_n-3)^2/2}}.$$

x_n	0	1	3	4
$\gamma(z_{n1})$	0.982	0.881	0.119	0.018
$\gamma(z_{n2})$	0.018	0.119	0.881	0.982

(e.g. for $x = 0$: $e^{-1/2} / (e^{-1/2} + e^{-9/2}) = 1 / (1 + e^{-4}) = 0.982$.)

M-step. $N_1 = N_2 = 2.0$, so $\pi_1 = \pi_2 = 0.5$, and

$$\mu_1 = \frac{1}{2} (0 \cdot 0.982 + 1 \cdot 0.881 + 3 \cdot 0.119 + 4 \cdot 0.018) = 0.655, \quad \mu_2 = 3.345 \text{ (by symmetry),}$$

$$\sigma_1^2 = \sigma_2^2 = \frac{1}{2} \sum_n \gamma(z_{n1})(x_n - 0.655)^2 = 0.691.$$

The new log-likelihood is $-6.462 > -7.158$: the means moved **outwards** toward their clumps and the variances **shrank**, and the likelihood went up, as the theory in Section 8 guarantees.

Key Rule: Initialization and Cost

- Run **K-means first** to find a suitable initialization for a GMM, which is then adapted using EM (e.g. μ_k from the centres, Σ_k from each cluster's sample covariance, π_k from the fraction of points in each cluster).
- EM takes **many more iterations** to reach (approximate) convergence than K-means,
- and each cycle requires **significantly more computation**.

Worked Example: K-Means as a Limit of EM (Scribe's Note)

Not from the slides; follows Bishop Sec. 9.3.2. Fix every covariance to $\Sigma_k = \epsilon \mathbf{I}$ with the same small ϵ . Then

$$\gamma(z_{nk}) = \frac{\pi_k \exp\{-\|\mathbf{x}_n - \mu_k\|^2/2\epsilon\}}{\sum_j \pi_j \exp\{-\|\mathbf{x}_n - \mu_j\|^2/2\epsilon\}} \xrightarrow{\epsilon \rightarrow 0} r_{nk},$$

because the term with the smallest distance dominates the denominator. The soft responsibilities become the **hard** assignments of k-means, and the M-step for μ_k becomes the k-means update. K-means is thus EM for a GMM with **equal, spherical, vanishing** covariances.

Summary: K-Means vs. EM for GMMs

	K-Means	EM for GMM
Model	none (distortion J)	probabilistic: $\sum_k \pi_k \mathcal{N}(\mathbf{x} \mid \mu_k, \Sigma_k)$
Assignment	hard $r_{nk} \in \{0, 1\}$ (closest centre)	soft $\gamma(z_{nk}) \in [0, 1]$ (posterior)
Parameters	centres μ_k	μ_k, Σ_k, π_k
Cluster shape	circles, equal size	ellipses, any size and weight
Objective	minimize J	maximize $\ln p(\mathbf{X} \mid \theta)$
Cost	few, cheap iterations	many, more expensive iterations

7 The General EM Algorithm

Definition : Log-Likelihood with Latent Variables

The goal of EM is to find **maximum likelihood solutions for models having latent variables**. The log-likelihood is

$$\underbrace{\ln p(\mathbf{X} \mid \boldsymbol{\theta}) = \ln \left\{ \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) \right\}}_{\text{discrete latent variable}} \quad \text{or} \quad \underbrace{\ln p(\mathbf{X} \mid \boldsymbol{\theta}) = \ln \left\{ \int_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) d\mathbf{Z} \right\}}_{\text{continuous latent variable}},$$

where $\mathbf{X}$ is the set of all **observed data** (the n th row is $\mathbf{x}_n^T$), $\mathbf{Z}$ the set of all **latent variables** (row $\mathbf{z}_n^T$), and $\boldsymbol{\theta}$ the set of all **model parameters**.

Key Difficulty: Why the Marginal Does Not Simplify

- The summation/integration over the latent variables **appears inside the logarithm**.
- The marginal $p(\mathbf{X} \mid \boldsymbol{\theta})$ **does not simplify** even if the joint $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$ belongs to the exponential family (e.g. Gaussian).
- This results in **complicated (non closed-form) expressions** for the maximum likelihood solution.

Definition : Complete vs. Incomplete Data

- **Complete data** $\{\mathbf{X}, \mathbf{Z}\}$: both the observation $\mathbf{X}$ and the latent variable $\mathbf{Z}$ are known (e.g. the cluster of each point is known – coloured points).
- **Incomplete data** $\mathbf{X}$: the observation is known, but the latent variable $\mathbf{Z}$ is unknown (all points the same colour).

Maximizing the **complete-data** log-likelihood $\ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$ is straightforward (Lecture 6). In practice we are only given $\mathbf{X}$, so instead we maximize the **expected value** of the complete-data log-likelihood under the posterior $p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta})$, and iterate.

Key Rule: E-Step and M-Step

Expectation step. Use the current parameters $\boldsymbol{\theta}^{\text{old}}$ to find the posterior of the latents $p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$, then use it to find the expectation of the complete-data log-likelihood for a general $\boldsymbol{\theta}$:

$$\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \sum_{\mathbf{Z}} p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = \mathbb{E}_{\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}} [\ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})].$$

Maximization step. Determine the revised estimate

$$\boldsymbol{\theta}^{\text{new}} = \arg \max_{\boldsymbol{\theta}} \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \arg \max_{\boldsymbol{\theta}} \sum_{\mathbf{Z}} p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \underbrace{\ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})}_{\text{log now inside the sum!}}$$

Since the logarithm now acts **directly on the joint** $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$, the M-step is in **closed form** (for exponential-family joints).

Algorithm: The General EM Algorithm

Given: a joint distribution $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$ over observed $\mathbf{X}$ and latent $\mathbf{Z}$, governed by $\boldsymbol{\theta}$. **Goal:** maximize the likelihood $p(\mathbf{X} \mid \boldsymbol{\theta})$ w.r.t. $\boldsymbol{\theta}$.

- 1: Choose an initial setting for the parameters $\boldsymbol{\theta}^{\text{old}}$
- 2: **repeat**
- 3: **E-step:** evaluate $p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$
- 4: **M-step:** evaluate $\boldsymbol{\theta}^{\text{new}} = \arg \max_{\boldsymbol{\theta}} \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}})$, $\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \sum_{\mathbf{Z}} p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$
- 5: Check for convergence of either the log-likelihood or the parameter values; if not converged, $\boldsymbol{\theta}^{\text{old}} \leftarrow \boldsymbol{\theta}^{\text{new}}$
- 6: **until** converged

Key Difficulty: Singularities, and MAP as a Remedy

A Gaussian component can **collapse onto a single data point**: if $\boldsymbol{\mu}_k = \mathbf{x}_n$ and $\boldsymbol{\Sigma}_k \rightarrow 0$, that component's density at $\mathbf{x}_n$ – and hence the likelihood – goes to **infinity**. Maximum likelihood is therefore ill-posed for GMMs, and EM can be drawn toward such degenerate solutions. The problem can be alleviated by applying **MAP** to $\mathcal{Q}$, i.e. maximizing

$$\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) + \ln p(\boldsymbol{\theta}),$$

with a **Dirichlet prior** on π_k and a **normal-inverse-gamma / normal-inverse-Wishart** (conjugate) prior on $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, as in Lecture 6. The prior penalizes vanishing variances, so the collapsed solution is no longer optimal. (*The slide words this as “normalized inversed Gaussian”; the intended prior is the conjugate normal-inverse-gamma of Lecture 6, or normal-inverse-Wishart in the multivariate case.*)

7.1 Gaussian Mixture Revisited**Worked Example: The Complete-Data Log-Likelihood of a GMM**

If $\mathbf{Z}$ were observed (both nodes shaded in the plate diagram),

$$\begin{aligned} \ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) &= \ln \prod_{n=1}^N \prod_{k=1}^K \pi_k^{z_{nk}} \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_{nk}} \\ &= \sum_{n=1}^N \sum_{k=1}^K z_{nk} \underbrace{\left\{ \ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}}_{\text{log inside the summation!}}. \end{aligned}$$

We can **switch the sums** to $\sum_k \sum_n$ and look only at the points associated with each component k : each $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is maximized **separately** (the Lecture 6 Gaussian MLE on the points of cluster k), and π_k via a Lagrange multiplier (the Lecture 6 categorical MLE, $\pi_k = N_k/N$ with hard counts).

Problem: the latent variables $\mathbf{Z}$ are **not observed**!

Worked Example: The Expected Complete-Data Log-Likelihood of a GMM

Instead, maximize its expectation under the latent-variable posterior:

$$\begin{aligned}
 \mathbb{E}_{\mathbf{Z}}[\ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})] &= \sum_{\mathbf{Z}} p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = \sum_n \sum_k p(z_{nk} \mid \mathbf{x}_n, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{x}_n, z_{nk} \mid \boldsymbol{\theta}) \\
 &= \sum_n \sum_k \frac{p(\mathbf{x}_n \mid z_{nk}, \boldsymbol{\theta}^{\text{old}}) p(z_{nk} \mid \boldsymbol{\theta}^{\text{old}})}{p(\mathbf{x}_n \mid \boldsymbol{\theta}^{\text{old}})} \ln p(\mathbf{x}_n, z_{nk} \mid \boldsymbol{\theta}) \\
 &\hspace{20em} \text{(Bayes' rule)} \\
 &= \sum_n \sum_k \underbrace{\frac{\mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k^{\text{old}}, \boldsymbol{\Sigma}_k^{\text{old}}) \pi_k^{\text{old}}}{\sum_j \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_j^{\text{old}}, \boldsymbol{\Sigma}_j^{\text{old}}) \pi_j^{\text{old}}}}_{\gamma(z_{nk}) - \text{responsibility that stays fixed}} \ln p(\mathbf{x}_n, z_{nk} \mid \boldsymbol{\theta}) \\
 &= \sum_n \sum_k \gamma(z_{nk}) [\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)].
 \end{aligned}$$

This is the complete-data log-likelihood with the unknown z_{nk} replaced by its **expectation** $\mathbb{E}[z_{nk}] = \gamma(z_{nk})$. As before we can **switch the sums**; since $\gamma(z_{nk})$ **stays fixed** during the M-step, each $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is maximized separately and π_k by a Lagrange multiplier – giving exactly the M-step updates of Section 6, now in closed form.

8 The Theory Behind the EM Algorithm

Maximizing $\ln p(\mathbf{X} \mid \boldsymbol{\theta}) = \ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})$ is difficult because of the marginalization inside the logarithm. It turns out that $\ln p(\mathbf{X} \mid \boldsymbol{\theta})$ can be maximized by **maximizing a lower bound** $\mathcal{L}(q, \boldsymbol{\theta})$ within the EM algorithm.

Definition : Lower Bound and KL Decomposition

For **any** distribution $q(\mathbf{Z})$ defined over the latent variables,

$$\ln p(\mathbf{X} \mid \boldsymbol{\theta}) = \underbrace{\mathcal{L}(q, \boldsymbol{\theta})}_{\text{lower bound}} + \underbrace{\text{KL}(q \parallel p)}_{\text{KL divergence}},$$

where

$$\mathcal{L}(q, \boldsymbol{\theta}) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})}{q(\mathbf{Z})} \right\}, \quad \text{KL}(q \parallel p) = - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta})}{q(\mathbf{Z})} \right\} \geq 0.$$

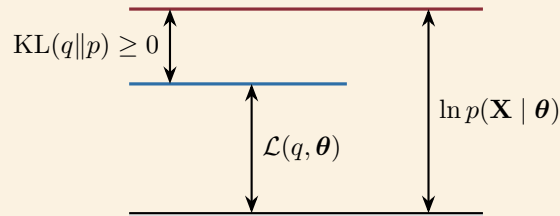
Proof: The Decomposition Holds for Any $q(\mathbf{Z})$

$$\begin{aligned} \mathcal{L}(q, \boldsymbol{\theta}) + \text{KL}(q \parallel p) &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})}{q(\mathbf{Z})} \right\} - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta})}{q(\mathbf{Z})} \right\} \\ &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \left\{ \underbrace{\ln p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta})}_{\ln p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}) + \ln p(\mathbf{X} \mid \boldsymbol{\theta})} - \ln q(\mathbf{Z}) - \ln p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}) + \ln q(\mathbf{Z}) \right\} \\ &= \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln p(\mathbf{X} \mid \boldsymbol{\theta}) = \ln p(\mathbf{X} \mid \boldsymbol{\theta}), \end{aligned}$$

using the product rule $p(\mathbf{X}, \mathbf{Z} \mid \boldsymbol{\theta}) = p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}) p(\mathbf{X} \mid \boldsymbol{\theta})$ and $\sum_{\mathbf{Z}} q(\mathbf{Z}) = 1$. ■

Key Rule: $\mathcal{L}(q, \boldsymbol{\theta})$ Is a Lower Bound

Because the Kullback-Leibler divergence satisfies $\text{KL}(q \parallel p) \geq 0$ (with equality iff $q(\mathbf{Z}) = p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta})$), the quantity $\mathcal{L}(q, \boldsymbol{\theta})$ is a **lower bound** on the log-likelihood $\ln p(\mathbf{X} \mid \boldsymbol{\theta})$, for every choice of q .



The decomposition $\ln p(\mathbf{X} \mid \boldsymbol{\theta}) = \mathcal{L}(q, \boldsymbol{\theta}) + \text{KL}(q \parallel p)$. (After Bishop.)

Key Rule: The E-Step Closes the Gap

Holding $\boldsymbol{\theta}^{\text{old}}$ fixed, the lower bound $\mathcal{L}(q, \boldsymbol{\theta}^{\text{old}})$ is **maximized w.r.t. q** by choosing $q(\mathbf{Z}) = p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$.

Proof. With this choice,

$$\text{KL}(q||p) = - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \underbrace{\left\{ \frac{p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}})}{p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}})} \right\}}_{=1, \ln=0} = 0,$$

and since $\ln p(\mathbf{X} | \boldsymbol{\theta}^{\text{old}})$ does not depend on q , $\mathcal{L}(q, \boldsymbol{\theta}^{\text{old}}) = \ln p(\mathbf{X} | \boldsymbol{\theta}^{\text{old}})$ is at its maximum: the bound **touches** the log-likelihood. $\square$

Equivalence with $\mathcal{Q}$. Substituting $q(\mathbf{Z}) = p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$,

$$\begin{aligned} \mathcal{L}(q, \boldsymbol{\theta}) &= \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z} | \boldsymbol{\theta}) - \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \\ &= \mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) + \text{const}, \end{aligned}$$

where the constant is the entropy of q , independent of $\boldsymbol{\theta}$. So **maximizing $\mathcal{Q}$ in the M-step is maximizing the lower bound**.

Key Rule: The M-Step Raises the Likelihood

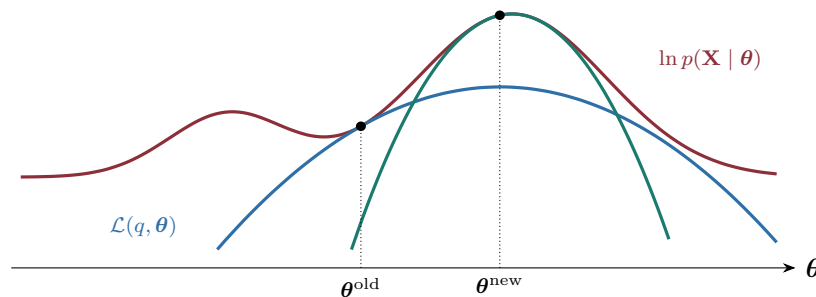
Now $q(\mathbf{Z})$ is **held fixed** and $\mathcal{L}(q, \boldsymbol{\theta})$ is maximized w.r.t. $\boldsymbol{\theta}$ to give $\boldsymbol{\theta}^{\text{new}}$:

$$\mathcal{L}(q, \boldsymbol{\theta}^{\text{new}}) = \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln p(\mathbf{X}, \mathbf{Z} | \boldsymbol{\theta}^{\text{new}}) + \text{const},$$

$$\text{KL}(q||p) = - \sum_{\mathbf{Z}} p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}}) \ln \left\{ \frac{p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{new}})}{p(\mathbf{Z} | \mathbf{X}, \boldsymbol{\theta}^{\text{old}})} \right\}.$$

The bound goes up, and since q was the **old** posterior, the KL term is now generally **positive** again. Because the KL divergence is nonnegative, the log-likelihood $\ln p(\mathbf{X} | \boldsymbol{\theta})$ **increases by at least as much as the lower bound does**:

$$\ln p(\mathbf{X} | \boldsymbol{\theta}^{\text{new}}) \geq \mathcal{L}(q, \boldsymbol{\theta}^{\text{new}}) \geq \mathcal{L}(q, \boldsymbol{\theta}^{\text{old}}) = \ln p(\mathbf{X} | \boldsymbol{\theta}^{\text{old}}).$$



(Schematic redrawn by the scribes after Bishop Fig. 9.14.) Blue: the bound built at $\boldsymbol{\theta}^{\text{old}}$ touches $\ln p(\mathbf{X} | \boldsymbol{\theta})$ there; its maximum gives $\boldsymbol{\theta}^{\text{new}}$. Teal: the next bound, which touches at $\boldsymbol{\theta}^{\text{new}}$.

Summary: EM as Coordinate Ascent on $\mathcal{L}(q, \theta)$

- **E-step:** compute the lower bound given the old parameters θ^{old} (blue curve) by setting $q(\mathbf{Z}) = p(\mathbf{Z} \mid \mathbf{X}, \theta^{\text{old}})$ – the bound touches $\ln p(\mathbf{X} \mid \theta)$ at θ^{old} .
- **M-step:** maximize this lower bound to get new parameters θ^{new} .
- This is **repeated** (green/teal curve) until convergence. Each iteration can **never decrease** the log-likelihood, so EM converges – but, like k-means, only to a **local** maximum (or saddle point), which is why initialization matters.

Scribe's note: the slide (following Bishop) calls the bound “convex”; for mixtures of exponential-family components it is a function with a **unique maximum**, i.e. concave in θ – “bowl facing down”, as drawn.

9 Summary

- Used the non-probabilistic k-means algorithm to solve the clustering problem.
- Described the Gaussian-mixture model.
- Applied the Expectation-Maximization algorithm for estimation of both the unknown parameters and latent variables.

Key Takeaways

1. Clustering is a **chicken-and-egg** problem: cluster parameters and data association are both unknown, and each is easy given the other – so we **alternate**.
2. **K-means** minimizes the distortion $J = \sum_n \sum_k r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$ by alternating **hard assignments** to the closest centre and **mean updates** $\boldsymbol{\mu}_k = \sum_n r_{nk} \mathbf{x}_n / \sum_n r_{nk}$; J never increases.
3. A **GMM** $p(\mathbf{x}) = \sum_k \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ is the marginal of the DGM $\mathbf{z} \rightarrow \mathbf{x}$ with a 1-of- K latent $\mathbf{z}$, $p(\mathbf{z}) = \prod_k \pi_k^{z_k}$. The **responsibility** $\gamma(z_k) = p(z_k = 1 \mid \mathbf{x})$ is the posterior, by Bayes' rule.
4. The GMM log-likelihood has a **sum inside the log**, so the ML equations $\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_n \gamma(z_{nk}) \mathbf{x}_n$, $\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_n \gamma(z_{nk}) (\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^T$, $\pi_k = N_k/N$ are **not closed-form** – γ depends on the parameters.
5. **EM** iterates: **E-step** computes $p(\mathbf{Z} \mid \mathbf{X}, \theta^{\text{old}})$ (for a GMM, the $N \times K$ table of responsibilities); **M-step** maximizes $\mathcal{Q}(\theta, \theta^{\text{old}}) = \mathbb{E}_{\mathbf{Z} \mid \mathbf{X}, \theta^{\text{old}}}[\ln p(\mathbf{X}, \mathbf{Z} \mid \theta)]$, where the log acts directly on the joint, so it is closed-form. Initialize with k-means.
6. ML for GMMs has **singularities** (a component collapsing onto one point); MAP with a Dirichlet prior on π_k and a conjugate prior on $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$ alleviates this.
7. **Theory:** $\ln p(\mathbf{X} \mid \theta) = \mathcal{L}(q, \theta) + \text{KL}(q \parallel p)$ for any q . The E-step sets $q = p(\mathbf{Z} \mid \mathbf{X}, \theta^{\text{old}})$ so $\text{KL} = 0$; the M-step raises $\mathcal{L}$, and the likelihood rises by at least as much. EM therefore **never decreases** the likelihood and converges to a local maximum.

Acknowledgement: Slide content adapted from C. Bishop, “Pattern Recognition and Machine Learning” (Ch. 8, 9); K. Murphy, “Machine Learning: A Probabilistic Perspective” (Ch. 11); M. I. Jordan, “An Introduction to Probabilistic Graphical Models” (Ch. 10, 11); D. Koller & N. Friedman, “Probabilistic Graphical Models” (Ch. 19.2.2); S. Prince, “Computer Vision: Models, Learning, and Inference” (Ch. 7.1–7.4, 7.8).